Measurement, missingness, and twin models: a whole-dissertation replication checkpoint of Schwabe (2016)

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Abstract

We partly replicate Schwabe's dissertation-level results: published floor-effect slopes of 1.05 for sum scores and .03 for item analysis become 1.095 and −.105 in one full-length fit; the missing-covariate precision ordering agrees in all 108 primary point contrasts. Whole-dissertation replication remains incomplete: empirical inputs are unavailable and production simulation counts unfinished. The item-slope sign reversal is inconclusive, with posterior interval $[-.698, .439]$. MAR heritability overestimation is not established: at five covariates and 10% missingness, the source gives .533 versus our .498 against truth .500; this opposite bias sign is also inconclusive. A published standard error of 0.04149 becomes 0.05907 on the variance scale, versus 0.04166 on the square-root path scale. The latter is an exploratory different-estimand reconstruction, not a reproduced procedure.

Claim	Published	Ours	Same direction?	Verdict
Floor spurious interaction: sum minus item slope	$1.05 - .03 = 1.02$	$1.095 - (-.105) = 1.200$; one dataset	Yes	partial
Missing-covariate precision: deletion minus full SE	Positive point contrasts	Positive at all 108 primary cells	Yes	partial
MAR deletion heritability bias, $q = 5$, 10% missing	$.533 - .500 = +.033$	$.498 - .500 = -.002$	No; inconclusive	partial
Empirical interaction findings, Chapters 3–5	Math $\beta_1 = 2.0837$; liberalism -2.81; SES $\beta_{1c} = -3.0777$	No raw-data refits	Unknown	not established
Missing-data standard-error magnitude, one example	0.04149	0.05907 (variance); 0.04166 (path)	Metric unresolved	partial

The source values are from Schwabe (2016, Tables 2.1, 3.2, 4.3, 5.4 and Figures 6.3 and 6.5). Unavailable empirical fits are not established, not contradicted. Rounding agreement requires the same estimand to round half up to the printed value; the figure has no printed decimal precision. Conditional bands use our Monte Carlo error plus extraction bounds and do not recover source uncertainty. Positive deletion-minus-full SE means greater precision with full information.

1 What replicated, and what remains unestablished

The measurement and missingness mechanisms receive partial support, while the dissertation's empirical conclusions remain unreplicated. The floor comparison gives a positive spurious sumscore slope and an item-level interval containing zero; the source's item-level point estimate is positive and ours negative, an inconclusive point reversal. The standard-error comparisons favor full information at every primary point contrast, but MAR heritability bias reverses point sign at five covariates and 10% missingness: .533 in the source versus .498 here, against truth .500. Our adjusted bias interval includes zero, so the source overestimation finding is not established by this checkpoint. Neither result establishes the whole dissertation: original repeated-simulation summaries, several analytical assertions, and all empirical refits remain open.

The dissertation asks how measurement and incomplete covariates change genetic and environmental inference from twin data. It was awarded as a University of Twente PhD on 24 March 2016, supervised by C.A.W. Glas with adviser Stéphanie M. van den Berg; the final published version has 154 numbered pages and 164 PDF pages. We read the full document, including all seven chapters and Appendices A–G. The 41-group inventory in Appendix A treats every substantive original result as in scope. Cited background models and future extensions are distinguished from completed original work.

The central decomposition separates additive genetic variance A, shared environmental variance C, and individual environmental variance E. Monozygotic twins share genetic effects; dizygotic twins have genetic correlation one half. Item-response models connect latent phenotypes to observed item responses. Chapters 2–5 let environmental or genetic variance depend on a genotype or measured moderator; Chapter 6 models covariates jointly to retain incomplete observations.

2 What we did ourselves, and what we took from others

Our contribution is an auditable reconstruction and a full coverage inventory, not new empirical twin data. Schwabe supplied the models, designs, reported results and original programs. We wrote the simulation generators, fitting wrappers, source-vector extraction, comparisons, diagnostics and report. Original unlicensed programs and the full dissertation remain local; only limited attributed source crops are included. The Chapter 4 publisher supplement was checked separately (Schwabe et al., 2016).

The statistical engines preserve the important model distinctions. R 4.5.0 (R Core Team, 2025) runs JAGS 4.3.2 through rjags 4-16 for the Bayesian item-level models (Plummer, 2024); coda 0.19-4.1 supplies posterior intervals and effective sample sizes (Plummer et al., 2006). OpenMx 2.21.13 fits the Chapter 6 Gaussian likelihoods (Neale et al., 2016). lavaan 0.6-19 provides a second baseline ACE fit (Rosseel, 2012). A covariance-only lavaan model would not reproduce the jointly moderated item likelihood, so that part explicitly uses the original JAGS engine family. The fitted likelihoods are implemented in `openmx_replication.R:15` (joint model), `openmx_replication.R:31` (conditional deletion model), and `jags_replication.R:24` (moderated item model).

The independent checks have limited independence. lavaan and OpenMx share one generated dataset and the ACE model; their baseline variance estimates differ by at most about 5×10^{-8}. SLSQP and CSOLNP share the Chapter 6 graph and data, checking optimization rather than model validity. Python's multivariate-normal generator checks R's conditional Gaussian generator through covariance moments. Original and reconstructed JAGS models share the engine and priors. Python 3.12.14 with NumPy 2.3.5, SciPy 1.17.0, Matplotlib 3.10.8 and PyMuPDF 1.26.6 supplies scalar checks, bands, plots and PDF inspection. R dependencies include MASS 7.3-65 for Gaussian generation, matrixStats 1.5.0 and foreign 0.8-90 for package replay. These checks do not provide independent empirical evidence; `provenance/independence.json` states each boundary.

3 Measurement effects: what the Bayesian attempts show

The full-length floor example reproduces the qualitative contrast but not the published repeated-simulation estimates. At 500 pairs and 60 items, our sumscore slope is 1.095 with posterior standard deviation .181; the item-level slope is −.105 with posterior standard deviation .297. Both use the same generated response matrix. The repaired original centred JAGS program gives an item-level slope of −.139, compared with −.105 from the new non-centred representation. All five monitored posterior means fall inside exploratory five-comparison Monte Carlo bands; the largest point potential scale-reduction factor is 1.010, with upper bound 1.031. Effective sample sizes remain finite, so this is implementation agreement rather than proof of convergence or a repeated-simulation result (`outputs/posterior-comparison.csv`). The source reported averages over 100 datasets, 1.05 and .03, so these are not like-for-like Monte Carlo means. The difference is evidence to investigate, not a rounding discrepancy.

The ceiling and symmetric sumscore fits supply supporting directional checks. Our ceiling slope is −.857, versus the reported −1.08; the symmetric case is .100, versus the reported mean .03 and between-replication standard deviation .24. Our posterior standard deviations are not substitutes for that between-replication standard deviation. Full results, intervals, effective sample sizes and chain lengths are in `outputs/confirmation-c2-floor.csv`, `outputs/confirmation-c2-floor-sums.csv`, `outputs/confirmation-c2-ceiling-sums.csv`, and `outputs/confirmation-c2-symmetric-sums.csv`.

Every published Bayesian simulation arm was attempted, and most remain unresolved. The twelve sample-size/effect cells of Tables 2.2–2.3 and all seven sample-size/item-count cells of Tables 5.1–5.2 have actual full-size-data pilot fits. Their short chains cannot establish power, parameter recovery, or convergence. The original curvilinear experiment on page 26 lacks a recoverable design and numerical results. Its absence is retained as a blocker, not converted to an excluded result. The source numerical targets are in `outputs/source-simulation-targets.csv` and `outputs/source-power-targets.csv`.

The production cost is substantial, rather than a reason to redefine the dissertation. At 8,000 pairs, a separate timing run used ten burn-in and ten retained iterations; the measured sampling cost projects to about 213 single-worker elapsed days for that condition's 250 source-length repetitions alone. This is a rough extrapolation, not a hard execution denial or a completed attempt at those repetitions. Initialization, adaptation, parallel scaling and mixing can change it. The production target remains open in `outputs/resource-projection.json`.

4 Missing covariates: direction and numerical scale

Full information reduces the estimated variance-component standard errors in all 108 primary point contrasts at this checkpoint. The complete grid includes three missingness mechanisms, four covariate counts and three stated missingness proportions; a fourth, 3%, tests the conflicting plot labels. There are 1920 method-by-dataset fits, with 20 repetitions per condition at the recorded checkpoint. The original used 1,000 per condition. The primary contrast is $\mathrm{SE}_{\mathrm{deletion}} - \mathrm{SE}_{\mathrm{full}}$; its sign is separate from an ours-minus-source reproduction deviation.

The mean contrast is simultaneously positive in 104 of the 108 primary comparisons under the stated approximation. We use paired differences across independently generated datasets, save two-standard-error intervals, and apply Bonferroni-adjusted Student intervals across the 108 primary contrasts (Dunn, 1961); the 36 extra 3% contrasts form a separate family. Small-sample normal-mean approximations and selection of assumptions after the initial pilots limit these bands. They are not equivalence tests. The complete records are `outputs/signed-contrasts.csv` and `provenance/uncertainty-plan.json`.

The plotted magnitudes remain unresolved on the printed variance scale. For example, with two covariates and low MCAR missingness, the source's additive-genetic standard error is approximately 0.04149, whereas ours is 0.05907. Across the 216 source SE values, 191 lie outside conditional adjusted bands for variance SEs. Those bands include our Monte Carlo error and a conservative quarter-PDF-point extraction bound, but omit unknown source Monte Carlo error; they cannot establish an author error or a complete two-sample disagreement.

An exploratory change of metric explains much of the difference. For a positive path a with variance $V = a^2$, the delta-method relation is $\mathrm{SE}(V) = 2|a|\mathrm{SE}(a)$. Reconstructing path SEs gives 0.04166 in the example, and 0 of the 216 path-scale comparisons lie outside the corresponding conditional bands. These are different estimands. Agreement with a transformed expression does not prove that it was the source procedure; the original plotting and standard-error extraction code would settle that question. The printed variance-scale comparison is preserved in `outputs/source-comparisons.csv`.

The extraction retains every source curve, not just the illustrative example. We read vector paths for Figures 6.3–6.6, calibrated their axes against tick labels, and checked rendered evidence. The CSV retains coordinates, drawing identifiers and extraction bounds: `outputs/source-figure-values.csv`. Source point orderings have unknown uncertainty. The table below concerns standard-error ordering only. The bias inventory is separate.

Condition	Source / ours contrast	Interpretation
All conditions	No point reversal	Both metrics and low-missingness readings retained in `outputs/signed-contrasts.csv`.

Figure 6.5's bias findings require a separate signed contrast: estimate minus generating truth, with positive meaning overestimation. All 72 MAR method/parameter/condition cells are in `outputs/bias-contrasts.csv`. There are 23 source-versus-ours point-sign reversals; 19 source signs exceed the extraction bound. None of these reversals has our Bonferroni-adjusted 72-comparison interval excluding zero. For deletion heritability at five covariates and 10% missingness, source .533154 and ours .497505 lie on opposite sides of truth .5. Our bias $-.002495$ has adjusted interval $[-.098004, .093014]$. This is an inconclusive point reversal, not evidence of an opposite established bias. The family was added during independent review and is exploratory; source Monte Carlo uncertainty remains unknown. `simulation/bias_contrasts.py` retains two-SE and adjusted intervals, extraction bounds and separate ours-minus-source deviations.

5 Assumptions

5.1 Regression coefficients and covariate means

The stated explained variance fixes a quadratic form, not the individual coefficients. We use equal positive coefficients satisfying $\beta^{\mathsf{T}}(\Sigma_W + \Sigma_B)\beta = .64$, yielding total variance 1.64 and explained fraction .39024. A distinct unequal-coefficient sensitivity retains the same explained variance. The main reconstruction estimates a mean for each covariate; the Appendix F script ties them together. We test the shared-mean restriction separately. Neither supplied choice establishes the missing original driver.

5.2 Low missingness and moderator allocation

The thesis gives 2% in the Chapter 6 design and captions but 3% in plots and parts of the discussion. Both are retained; neither is silently selected because it agrees better. For Chapter 5 we assume a balanced binary moderator, consistent with a .7 main effect explaining approximately 11% of variance. The exact moderator allocation and random seeds were not recovered. The alternative sample-size reading of 80,000 in the prose is not a completed arm: Table 5.1 and its discussion specify 8,000, which is the implemented primary reading.

5.3 Priors, initialization, and historical software

The reconstructed Bayesian priors follow the source precision/variance conventions. The current and pre-defense BayesTwin repositories were both tested; an exact source-era session specification remains unavailable. New random seeds and a non-centred parameterization are explicit departures from unavailable original drivers. They must not be used to claim identical stochastic output. The full-length original-program comparison tests the same posterior on the same generated input, while source-length repeated grids remain open.

6 Ambiguities that were resolved locally

6.1 Program identifiers and modern R conditions

Appendix A fails with its documented inputs because it references `ndz`, `n.items` and `Ydz` instead of the documented names. Repairing only those identifiers allows sampling. Appendices B, D and E compile without those repairs, and Appendix G samples after local PDF whitespace/page-label normalization. These are executable checks, not empirical validation: `outputs/original-programs.csv` and `outputs/appendix-g-result.txt`.

The current BayesTwin code encounters a vector-condition error under R 4.5.0. Applying an explicit scalar test to the initialization check permits all twelve model/item-family smoke tests to return. The pre-defense revision has separate failures, including inconsistent-category initialization and a missing ADE prior argument. These are recorded alongside successes in `outputs/bayestwin-checks.csv` and `outputs/historical-bayestwin.csv`; the later package (Schwabe, 2017) is not substituted silently for the 2016 source.

6.2 Printed probability and information expressions

Two literal printed expressions do not match the implemented models under simple checks. At zero trait and thresholds with unit discrimination, Equations 4.6–4.9 give probabilities summing to 1.25; the Appendix B cumulative-softmax program gives probabilities summing to one. For Equation 3.4, a discrimination of two at response probability one half gives printed information .25, while finite-difference score information is 1.00. The usual discrimination-squared factor explains this difference. We checked the rendered expressions in `evidence/eq4-probabilities.png` and `evidence/eq3-information.png`. Neither check establishes that the empirical fitted programs or Figure 3.2 used those literal expressions.

6.3 Identification versus likelihood uniqueness

The binary-moderator sign ambiguity can be checked directly. Paths $(b_2, b_3) = (.5, .2)$ and $(.5, -1.2)$ give the same endpoint variances .25 and .49. Positive loglinear variances uniquely determine the intercept and slope at those two endpoints. This finite check supports the stated parameterization distinction, but does not prove a unique global likelihood maximum or the unrestricted continuous-moderator extension. Those broader assertions remain unresolved; `outputs/analytic-checks.json` records the check and its limits.

7 Empirical results and summary reconstructions

The original empirical conclusions cannot be refitted from the obtained materials. Chapter 3 requires NTR responses and Cito calibration; Chapter 4 requires Virginia30K responses; Chapter 5 additionally requires linked SES; Chapter 6 requires linked school and twin records. All twelve public author repositories were cloned and audited before these gaps were declared. The available BayesTwin example is explicitly simulated output, not a substitute empirical dataset.

Published-mean calculations provide limited checks with a different scope. The Chapter 3 one-standard-deviation environmental ratios reconstruct as 1.6316 and 1.8489, rounding to 1.63 and 1.85. A ratio of the printed variance means gives heritability .7292, whereas the reported posterior mean is .7286; these need not be equal because expectation and nonlinear division do not commute. The same distinction applies to Chapter 4. The sign reversals between its item and sumscore analyses are source-reported findings, not fresh empirical replications.

The illustrative figures can be partly regenerated without raw data. We simulated 10,000 MZ pairs from the Chapter 3 printed mean parameters, reconstructed conditional variance curves, and generated both Chapter 7 genetic/phenotypic illustrations. Original posterior bands and score histograms remain blocked. See `outputs/ch3-mz-illustration.pdf`, `outputs/conditional-variance.pdf` and `outputs/ch7-illustration.pdf`.

Claim	Published	Ours	Precision	Numerical	Scope
Ch6 MCAR,q2,low missing; A variance SE	0.04149	0.05907	Figure-derived	Not judged	Limited
Same condition, path SE	Metric not stated	0.04166	Figure-derived	Not comparable	Limited
Ch2 floor beta1: sum / IRT	1.05 / .03	1.095 / -.105	.01 in source	Not comparable	Limited; one dataset
Ch2 ceiling beta1, sums	-1.08	-.857	.01 in source	Not comparable	Limited; one dataset
Ch2 symmetric beta1, sums	.03	.100	.01 in source	Not comparable	Limited; one dataset
Ch3 AxE one-SD ratio	1.63	1.6316	.01	match	Limited
Ch3 AxC one-SD ratio	1.85	1.8489	.01	match	Limited
Ch3–6 empirical fitted results	Original tables	Not estimated	Table-specific	No verdict	Unavailable

The first two rows separate a labelled variance SE from an exploratory path SE; neither inherits a decimal tolerance from a figure-derived display. The next three rows compare a single dataset with a source Monte Carlo average and therefore have no numerical verdict. The ratio rows use the source

formula and round to the printed values, but retain limited scope. Empirical fits have unavailable scope. Per-family comparability records are in provenance/comparability.json.

8 What remains unresolved

The largest remaining causes differ by result, and the stopping statements preserve that distinction. For empirical claims, missing raw responses, calibration and linkage inputs rank ahead of software choices; repository and supplement searches did not recover them. For Bayesian simulation summaries, insufficient repeated converged fits rank ahead of numerical discrepancies; all cells were piloted, source-length cases and original-program checks were attempted, and actual resource costs were measured. The exact curvilinear design remains unavailable.

For the Chapter 6 magnitude gap, the standard-error estimand currently ranks ahead of optimizer differences. We checked vector extraction against the ink, retained variance versus path metrics, compared SLSQP with CSOLNP, checked covariance algebra independently, and ran unequal-coefficient and shared-mean sensitivities. These checks do not recover the original random seeds, beta allocation, session versions or plotting driver. The most informative single artifact would be the original Chapter 6 simulation-and-summary script. No author contact was made.

Independent scientific review identified missing bias contrasts and continuation bookkeeping defects; these have been corrected while preserving the original numerical baseline. Independent rendered-manuscript review checks the final PDF, source hashes and evidence in provenance/review.json. This review does not establish successful whole-dissertation replication. Completion of execution attempts is also distinct from successful replication. None of the open production or empirical items is closed as outside a selected scope.

9 Code and source availability

The dated availability audit below is part of the evidence, not a claim that every possible repository was exhaustively searched. Institutional metadata and the author-hosted source identify the edition; source hashes and repository revisions are in provenance/source.json and provenance/author-repositories.json. Full copyrighted source files remain local. The following rows copy provenance/availability.json.

Place/date	Result
Institutional final-version record / 2026-10-30	Verified PhD, institution, advisers, date, DOI, ISBN and final-version designation.
Author dissertation GitHub repository / 2026-10-30	Full164-page thesis and five model programs obtained at6c2c48901209fc76ba932bb094c61abba5ef160d. No raw empirical datasets or Monte Carlo driver/seeds. Full text read. Original files kept local; rights reserved/no code license.
All12 public author GitHub repositories / 2026-10-30	All cloned; immutable hashes in provenance/author-repositories.json. Data-file and code audit found workshop data, BayesTwin simulated posterior example and unrelated projects; no matching NTR/Cito/Virginia study inputs or curvilinear simulation driver. This is a search limit, not proof of absence.
OSF indexed search / 2026-10-30	Queries site:osf.io Schwabe twin and exact Inga Schwabe2016 found later unrelated projects/preprints, no matching dissertation simulation or empirical input. No identified thesis OSF project.

Place/date	Result
GitLab indexed search / 2026-10-30	Queries Schwabe and exact ingaschwabe returned unrelated authors/no matching repository. No claim of exhaustive private-site search.
Zenodo indexed search / 2026-10-30	Queries Schwabe twin and exact Inga Schwabe returned unrelated records/later projects; no matching dissertation data/code record.
Chapter4 article and supplement / 2026-10-30	Publisher confirms CC BY4.0 article and links ten-page supplement. Supplement accessible through browser; contains category plots, fit plots and JAGS program, not raw responses. Original thesis code replayed locally. Supplement findings cannot be refitted without Virginia30K responses.
BayesTwin repository history / 2026-10-30	Current code656afd72d45ecd13f54b72c9400cceface8d423f, version1.0 dated2017, GPL¿=2. Fetched full history; last pre-defense revision a8e5d6b5ee6a8c76bdd0234a00fbcc809712ac08 dated2015-10-15 available for separate replay. Example results.rda explicitly simulated, not dissertation empirical data.
Historical statistical engine specification / 2026-10-30	Thesis cites R/rjags2013 and OpenMx2011/JAGS2003 publications but no complete sessionInfo, engine patch versions, random seeds or driver. Exact historical environment not determined. Current pinned versions and original-model replay used with explicit scope limit.
Rejected candidate acquisition / 2026-10-30	Shell download attempt denied HTTP CONNECT403 by environment network policy. Candidate rejected; no retry or bypass. Schwabe author-hosted repository independently accessible.

10 Compliance ledger

The ledger preserves departures rather than converting an unfinished checkpoint into completion. The following rows copy `provenance/compliance.json`; justifications describe the current evidence boundary.

Requirement/status	Evidence and justification
Whole dissertation is the unit / followed	`coverage.csv` inventories all original result groups in Chapters1-7 and AppendicesA-G; background and future proposals distinguished in `provenance/source.json`. Feasible production work remains open, never out of scope.
Original code and source audit / followed	`provenance/availability.json` and `provenance/author-repositories.json`; all12 author repositories cloned; OSF/GitLab/Zenodo indexed searches and publisher supplement checked.
Preferred engine preflight / followed	`provenance/preflight.json` passes R4.5.0, lavaan0.6.19, OpenMx2.21.13; local package hashes recorded.
JAGS for Chapters2-5 / departed	`outputs/original-programs.csv`; `outputs/engine-agreement.csv`; `outputs/analytic-checks.json` The thesis jointly fits non-Gaussian item likelihoods with latent variance moderation in JAGS. A covariance-only lavaan fit would change the statistical model. JAGS4.3.2/rjags4-16 used; original programs replayed, independent noncentred generator/covariance checks and lavaan/OpenMx baseline provided. These do not independently validate the full posterior.

Requirement/status	Evidence and justification
Original Monte Carlo repetition and chain counts / departed	outputs/pilot-jags-planned.json; outputs/pilot-openmx-planned.json; outputs/timing-c5-largest-timing.csv; coverage.csv All stated conditions attempted with short diagnostics; a full-length Chapter2 case and expanded Chapter6 runs retained separately. Production counts are unresolved, not waived: the N8000 Chapter5 timing pilot projects months of single-worker elapsed computation for that condition alone.
Signed contrasts and two-axis decisions / followed	provenance/uncertainty-plan.json; Chapter6 deletion-minus-full SE contrast positive means smaller full-information SE. Source reproduction deviation is separate. Source rounded-summary reconstructions are not posterior refits. Bias contrasts added during review in outputs/bias-contrasts.csv; exploratory72-comparison family preserves23 inconclusive point reversals.
Approved manuscript layout / followed	Approved preamble at baseline e2b553c06c87; full revision in provenance/delivery.json. Semantic subject changes only.
Independent scientific and rendered-manuscript review / followed	provenance/scientific-review.json; provenance/rendered-review.json; provenance/review.json Separate reviewers inspected scientific evidence and rendered pages. Initial findings required bias-inventory, continuation and nonexistent-PR wording fixes; final hash-matched review records govern acceptance. Review does not close unreplicated results.
Scope and licensing / followed	Source PDF/unlicensed original programs kept local. Only original wrappers, results and limited attributed scholarly evidence crops included. No source-policy denial bypass; no author contact, merge, social post or reusable skill edits.
Runnable immutable version and remote byte verification / followed	README.md reproduction recipe; provenance/delivery.json records immutable code/artifact revisions and remote downloads when uploaded. Master navigation links await merge.

11 Next steps

The next checks require specific missing artifacts or production runs. Obtain the original Chapter 6 driver through a permitted public source, then test the standard-error metric and exact missingness design. Continue the Bayesian grid with source-length chains and independent convergence checks, using measured costs and checkpoints. Recover authorized empirical response/calibration/linkage inputs before claiming any empirical replication. The remaining feasible work and measured resource needs are enumerated in provenance/unfinished-work.json and outputs/production-resource-plan.json. Review of this checkpoint does not close those items.

A Whole-dissertation coverage inventory

Every listed result remains in scope. "Attempted/unresolved" includes genuine diagnostic or reconstruction work whose evidence does not establish the entire result group; "blocked" identifies missing inputs or an undocumented design after an actual acquisition audit. No successful smoke test closes a broader empirical claim.

Claim and source locator	Current evidence and remaining work
C1-1; Empirical; Fig1.2 p5. Three motivation-scale distributions; N4220; skewness -0.55,-0.19,-2.62	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C1-2; Illustration; Fig1.3 p8. Two logistic item characteristic curves	attempted/unresolved. Qualitative logistic ordering verified; exact source item difficulties and random seed are unreported. Next: Exact curve coordinates remain unresolved. Evidence: outputs/analytic-checks.json
C2-1; Analytical; Eqs2.1-2.16 pp14-19. ACE Rasch interaction construction; DZ covariance; positive logvariance; delta transformation	attempted/unresolved. DZ covariance and delta identities checked; baseline lavaan/OpenMx estimates agree. Next: Full moderated posterior validation remains separate. Evidence: outputs/analytic-checks.json; outputs/engine-agreement.csv
C2-2; Headline numerical; Table2.1 Fig2.1 pp21-22. Floor and ceiling: 100 replications each; all A C E beta estimates SDs posterior SDs; reliability skewness and score distributions	attempted/unresolved. Full-length one-dataset floor comparison and ceiling sumscore fit; both item/sum arms attempted in pilot. Original100-replicate summaries not established. Next: Run full replicated floor/ceiling comparisons and diagnose independent chains. Evidence: outputs/confirmation-c2-floor.csv; outputs/confirmation-c2-floor-sums.csv; outputs/confirmation-c2-ceiling-sums.csv; outputs/pilot-jags-planned.json
C2-3; Headline numerical; Tables2.2-2.3 pp22-23. 12 N-by-delta conditions; beta1 detection power; all four parameter mean SD posteriorSD; A C E detection power	attempted/unresolved. All12 N-by-delta cells executed. Short-chain power and recovery estimates are not valid production estimates. Next: Continue source-length chains and100 independent datasets per condition. Evidence: outputs/pilot-jags-planned.json; outputs/source-power-targets.csv; outputs/source-simulation-targets.csv
C2-4; Numerical; p24. Symmetric difficulty sumscore simulation: beta1 mean .03 SD .24	attempted/unresolved. One full-length symmetric case; source100-replicate mean .03/SD.24 not reproduced as an aggregate. Next: Complete original100 datasets; preserve all trials. Evidence: outputs/confirmation-c2-symmetric-sums.csv
C2-5; Numerical; p26. Curvilinear third simulation: satisfactory power but biased estimation	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C2-6; Analytical/software; pp16,25-27. 2PL and polytomous extension; unknown item calibration; computational advantage over comparison model	attempted/unresolved. 1PL/2PL/PCM/GPCM generators and fits attempted; current package works after explicit vector-condition repair. Original comparative speed benchmark not recoverable. Next: Production posterior validation and original timing comparison driver needed. Evidence: outputs/bayestwin-checks.csv; outputs/historical-bayestwin.csv
C3-1; Empirical; pp34-36. Sample construction; zygosity and missingness counts; Cito OPLM calibration linkage	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C3-2; Analytical; Eqs3.1-3.19 pp36-40. ACE/ADE and A-by-C/E interaction construction; test information identity	attempted/unresolved. Baseline/model covariance checked. Printed item-information expression differs from finite-difference score information when discrimination differs from1. Next: Actual test-information code and fixed item calibration required to establish effect on published curves. Evidence: outputs/analytic-checks.json; evidence/eq3-information.png
C3-3; Headline empirical; Table3.1 p41. Eight model DIC comparison; both-interaction ACE preferred	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json

Claim and source locator	Current evidence and remaining work
C3-4; Headline empirical; Table3.2 pp42-43. Six posterior summaries; positive AxE HPD; AxC HPD includes zero; h2; one-SD ratios 1.63 and1.85	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C3-5; Numerical illustration; Fig3.1 p43. 10k-twin simulated mean curves from fitted parameters; credible bands	attempted/unresolved. 10,000 MZ pairs generated at printed mean parameters; conditional curves reconstructed. Next: Joint posterior required for original95% credibility band; source seed absent. Evidence: outputs/ch3-mz-illustration.pdf; outputs/conditional-variance.pdf
C3-6; Empirical; Fig3.2 p43. Nine test-version information and standard error curves	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C3-7; Empirical; Fig3.3 p44. Score histogram; 35 perfect; 828 above mode52	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C4-1; Empirical; pp51-54. Virginia sample descriptors; reliability; item-total correlations; recoding	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C4-2; Headline empirical; Figs4.1-4.4 pp58-61. Homals eigenvalues/loadings/category positions; two-factor interpretation; frequency-loading regressions	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C4-3; Empirical; Table4.1 p62. Nine-item GPCM parameters chi-squares; alpha/lambda2; GPCM versus PCM AIC/BIC	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C4-4; Analytical/software; Eqs4.1-4.9 pp54-57; AppB. Joint GPCM genotype-environment model and category probabilities	attempted/unresolved. Original normalized GPCM program compiles/samples. Literal printed probability expressions sum1.25 at zero inputs, while code softmax sums1. Next: Empirical posterior validation requires raw item data; no inference of defective fitted program. Evidence: outputs/original-programs.csv; outputs/analytic-checks.json; evidence/eq4-probabilities.png
C4-5; Headline empirical; Tables4.2-4.4 pp63-64. Four DICs; item and sumscore posterior summaries and sign reversals; heritability	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C4-6; Empirical/illustration; Figs4.5-4.6 pp64-65. Conditional variance curves and uncertainty; sumscore histogram	attempted/unresolved. Conditional mean-parameter curves and signed published slope contrasts calculated. Next: Histogram and joint-posterior bands blocked by missing raw responses/draws. Evidence: outputs/conditional-variance.pdf; outputs/analytic-checks.json
C4-7; Empirical/software; p54 supplementary material. All-item category-position plots and all/twin-only fit plots; accompanying program	attempted/unresolved. Publisher supplement retrieved/read: category and fit plots, model program; no raw responses. Next: Refits blocked by missing Virginia30K responses; do not infer fit from plot-only reading. Evidence: provenance/availability.json; outputs/original-programs.csv

Claim and source locator	Current evidence and remaining work
C5-1; Headline analytical; Eqs5.3-5.15 pp74-77; AppC pp138-139. Logvariance ACE-by-M identification; squared-path sign ambiguity; monotonicity	attempted/unresolved. Binary-moderator sign ambiguity verified by two equal-variance parameterizations; loglinear endpoint mapping checked. Next: Global likelihood uniqueness and continuous-moderator extension not established by this finite check. Evidence: outputs/analytic-checks.json
C5-2; Headline numerical; Table5.1 pp78-79. 250 replicates each N1000/2000/4000/8000; seven parameters mean SD posteriorSD; alpha	attempted/unresolved. All four sample sizes executed including8000; pilot is unconverged and not a table reproduction. Next: Original250replicates x80000iterations retained as production target; projected cost recorded. Evidence: outputs/pilot-jags-planned.json; outputs/timing-c5-largest-timing.csv; outputs/source-simulation-targets.csv
C5-3; Headline numerical; Table5.2 pp79-80. 250 replicates each J20/100/250 at N1000; seven parameters mean SD posteriorSD; alpha	attempted/unresolved. All20/100/250-item arms executed atN1000; source-length recovery not established. Next: Continue original250replicates and diagnose mixing. Evidence: outputs/pilot-jags-planned.json; outputs/source-simulation-targets.csv
C5-4; Empirical; Table5.3 pp80-81. SES construction counts and longitudinal correlation .72	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C5-5; Headline empirical; Table5.4 Fig5.2 pp82-83. Eight posterior summaries; negative C-by-SES; histogram of b1c; other interactions inconclusive	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C5-6; Empirical; p83. C-only versus both-interaction empirical analysis changes direction	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C5-7; Software; p84 AppE pp143-146. Separate-twin moderator extension with cotwin correction and zygosity-specific regression	attempted/unresolved. Separate-moderator original program compiled and sampled synthetic data. Next: Full parameter-recovery study not supplied by thesis; smoke test does not establish empirical validity. Evidence: outputs/original-programs.csv
C6-1; Headline analytical; Eq6.5 pp89-91. Full-information joint covariance construction and missing-data likelihood	attempted/unresolved. Joint covariance independently constructed by linear transformation; optimizer check on same graph/data. Next: Empirical validity remains data-dependent. Evidence: outputs/analytic-checks.json; outputs/check-c6-CSOLNP-full.csv
C6-2; Headline numerical; Figs6.3-6.4 pp93-94. MCAR/MAR SEs for A C E: q2/3/4/5 by missing .02/.06/.10; deletion versus fullinfo	attempted/unresolved. All MCAR/MAR conditions attempted and checkpointed, including2/3percent ambiguity; variance and path-SE metrics kept separate. Next: Complete1000source-count repetitions and resolve original SE extraction procedure. Evidence: outputs/trial-records.csv; outputs/source-comparisons.csv; outputs/signed-contrasts.csv
C6-3; Headline numerical; Fig6.5 p94. MAR bias in A C and h2 for deletion; all12conditions	attempted/unresolved. All72 MAR signed bias cells inventoried;23 source-versus-ours point reversals, none separated from zero by our adjusted bands; production and source-driver gaps remain. Next: Complete source-count Monte Carlo and identify original regression coefficients. Evidence: outputs/trial-records.csv; outputs/source-comparisons.csv; outputs/bias-contrasts.csv
C6-4; Headline numerical; Fig6.6 p95. MNAR SEs; all12conditions and bothmethods	attempted/unresolved. All MNAR conditions fitted, retaining convergence failures and signed comparisons. Next: Complete source-count Monte Carlo; original missingness generator/SE procedure needed. Evidence: outputs/trial-records.csv; outputs/source-comparisons.csv; outputs/signed-contrasts.csv

Claim and source locator	Current evidence and remaining work
C6-5; Numerical; pp92-96 untabulated. Beta recovery; MCAR all components and h2; MAR E recovery; MNAR near truth with E .21at10%	attempted/unresolved. Untabulated regression/component means included for every condition; no selection by agreement. Next: Complete original repetitions; original untabulated outputs would settle comparison. Evidence: outputs/condition-summary.csv
C6-6; Headline empirical; Tables6.1-6.3 pp98-100. 990pair descriptors missingness; ACE h2;13coefficients 99.6%HPD allinclude0	blocked. Inputs not recovered from full author-repository audit; not evidence of nonexistence. Next: Obtain stated input through permitted public source; no author contact authorized. Evidence: provenance/author-repositories.json; provenance/availability.json
C6-7; Software; AppF pp147-149. OpenMx full-information original script	attempted/unresolved. Original OpenMx model ran with q2 andq5 after omitting only install/source setup; also tested dynamic label variant. Next: Historical engine/sessionInfo and original driver unavailable. Evidence: outputs/original-openmx.csv
C6-8; Software; AppG pp150-154. Bayesian ordinal-liability extension with uncertain links	attempted/unresolved. PDF program normalized locally, compiled and sampled with missing synthetic binary/continuous covariates. Next: Empirical table6.3 still blocked; no convergence claim from smoke test. Evidence: outputs/appendix-g-result.txt; provenance/appendix-g-extraction.json
C7-1; Illustration; Fig7.1 p110. 10k genetic/phenotypic scatter: h2=1 and .5	attempted/unresolved. Both10,000-individual illustrations regenerated with theoretical correlations1 andsqrt(.5). Next: Exact source points require original seed; mechanism checked. Evidence: outputs/ch7-illustration.pdf; outputs/analytic-checks.json
C7-2; Software; p114. BayesTwin available implementation of most discussed models	attempted/unresolved. All12 ACE/ADE/AE x1PL/2PL/PCM/GPCM cases attempted in current and pre-defense source. Historical failures and modern vector-condition repair retained. Next: Broader production validation and original engine dependencies remain unresolved. Evidence: outputs/bayestwin-checks.csv; outputs/historical-bayestwin.csv
A-1; Software; pp131-133. AE_1PL original program compiles and estimates ACE-by-E despite filename	attempted/unresolved. Three identifier repairs permit source-length original JAGS fit. Five means agree with independent parameterization within exploratory conditional MC bands; largest point Rhat1.010, upperbound1.031. Next: More-chain/tail convergence checks and original100-dataset production; not closed by one dataset. Evidence: outputs/original-programs.csv; outputs/posterior-comparison.csv
D-1; Software; pp140-142. ACE_M original program compiles and estimates shared-moderator model	attempted/unresolved. Unmodified shared-moderator program compiled and sampled synthetic data. Next: Production recovery is tracked in C5-2/C5-3, not claimed by compilation. Evidence: outputs/original-programs.csv

B Artifacts and reproduction

The runnable recipe is README.md, sections "Re-extract source targets and replay software", "Rerun simulations and resume checkpoints", and "Rebuild the report from saved outputs". The historical baseline is distinct from the immutable runnable code revision in provenance/delivery.json. Rebuilding this manuscript from saved outputs does not rerun simulations. Master links below are intended navigation and await merge; provenance/delivery.json supplies commit-pinned review links. PR creation remains denied; no PR exists for this checkpoint.

Claim/evidence	File and regeneration entry
Whole-dissertation accounting	coverage.csv; audited against the entire source.
Bayesian grids and source targets	simulation/jags_replication.R; simulation/run_grid.py; simulation/extract_tables.py.
Missing-data fits and all trial results	simulation/openmx_replication.R; outputs/trial-records.csv; simulation/summarize.py.
Signed bias and resource evidence	outputs/bias-contrasts.csv; simulation/bias_contrasts.py; simulation/resource_plan.py.
Signed contrasts and source comparisons	outputs/signed-contrasts.csv; outputs/source-comparisons.csv.
Vector extraction and evidence	simulation/extract_figures.py; provenance/figure-extraction.json; provenance/source-crops.json.
Original program and historical package checks	simulation/original_programs.R; simulation/original_confirmation.R; simulation/appendix_g_check.R; simulation/bayestwin_check.R; simulation/historical_bayestwin.R.
Analytical and engine checks	simulation/analytic_checks.py; simulation/engine_checks.R; outputs/engine-agreement.csv.
Editable manuscript and tables	paper/paper.tex; paper/references.bib; simulation/build_report.py.

C Selected source evidence

These limited attributed crops show the printed expressions and standard-error labels discussed above. Copyright remains with Schwabe (2016). Full-resolution evidence and the remaining crops are linked in provenance/source-crops.json; the full dissertation is not redistributed.

transformed into item information curves. For a dichotomous item, the item information curve for item k can be calculated as follows:

$$I_k(\theta) = P_k(\theta)(1 - P_k(\theta)) \tag{3.4}$$

where $P_k(\theta)$ refers to the conditional probability of answering item k correctly (i.e, the IRC) given θ and item parameters a_k and b_k (Embretson & Reise, 2009). Under the assumption that conditional on θ, the item responses are uncorrelated (i.e., local independence), item information

$$p(y_{ijk} = \text{“no”}|\theta_{ij}, \alpha_k, \beta_{kl}) = \left[1 + e^{\alpha_k(\theta_{ij} - \beta_{k1})} e^{\alpha_k(\theta_{ij} - \beta_{k2})}\right]^{-1} \tag{4.6}$$

$$p(y_{ijk} = \text{“?”}|\theta_{ij}, \alpha_k, \beta_{kl}) = p(y_{ijk} = \text{“no”}|.) \times e^{\alpha_k(\theta_{ij} - \beta_{k1})} \tag{4.7}$$

$$p(y_{ijk} = \text{“yes”}|\theta_{ij}, \alpha_k, \beta_{kl}) = p(y_{ijk} = \text{“no”}|.) \times p(y_{ijk} = \text{“?”}|.) \tag{4.8}$$

$$\times e^{\alpha_k(\theta_{ij} - \beta_{k2})} \tag{4.9}$$

Figure 1: Schwabe (2016), printed Equation 3.4 and Equations 4.6–4.9. Literal-expression checks do not establish which expressions the empirical analysis used.

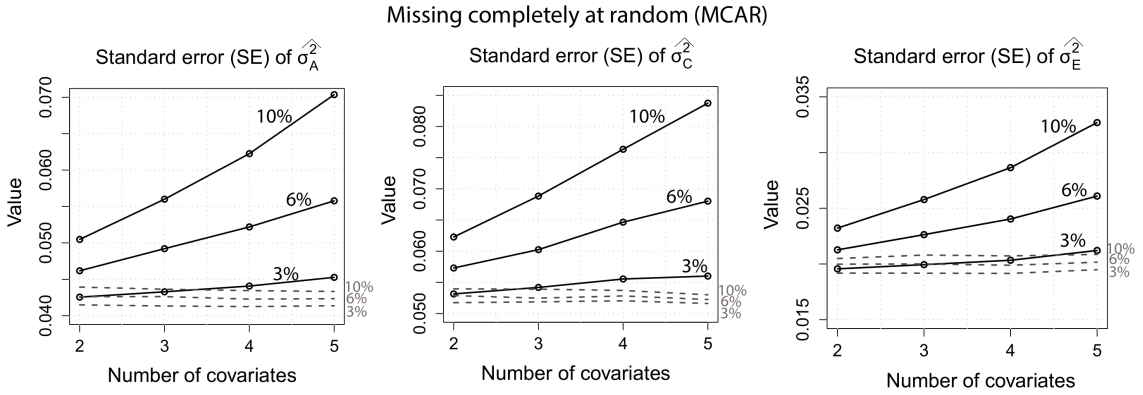


Figure 2: Schwabe (2016), Figure 6.3, printed page 93. The source labels variance-component standard errors; reconstruction of path standard errors remains a distinct, exploratory hypothesis.

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